

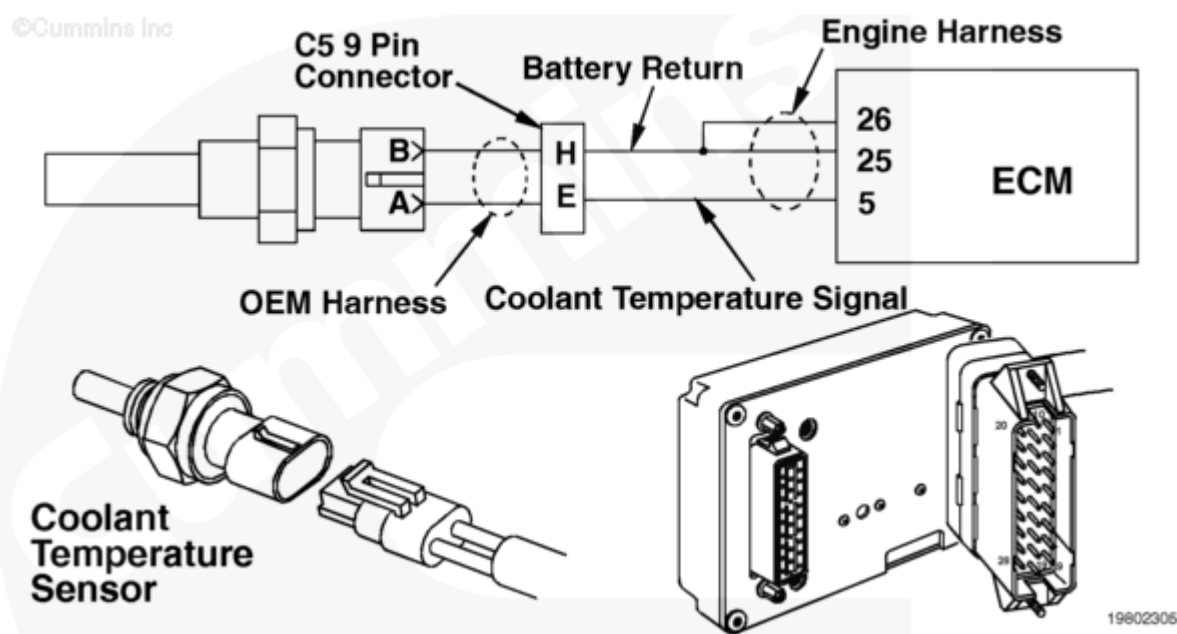
Fault Code: 145

Engine Coolant Temperature 1 Sensor Circuit - Voltage Below Normal or Shorted to Low Source

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 145 PID(P): 110 SPN: 110 FMI: 16 Lamp: On SRT:	Low signal voltage detected at the engine coolant temperature sensor circuit.	None on performance.



Coolant Temperature Sensor Circuit

Circuit Description

The engine coolant temperature sensor is a variable resistor sensor used by the ECM to monitor the engine coolant temperature. Signals from the engine coolant temperature sensor are not used by the CENTRY™ ECM to control the engine. The engine coolant temperature sensor has two circuits: signal and return circuits. The signal voltage indicates the coolant temperature.

Component Location

The location of the coolant temperature sensor may vary and is OEM dependent.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected that the engine coolant temperature sensor signal voltage was out of range low.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light when the diagnostic runs and fails.
- No engine protection for engine coolant temperature.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- No engine protection for engine coolant temperature.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Malfunctioning or damaged OEM wiring harness.
- Damaged or loose connectors.
- Malfunctioning or damaged engine coolant temperature sensor.

Refer to Troubleshooting Fault Code t05-145 (/qs3/pubsys2/xml/en/procedures/98/98-t05-145.html)

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